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B. E. (Eighth Semester) Examination, April-May, 2019

(New Scheme)

(Mech. Engg. Branch)

FINITE ELEMENT METHODS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) is compulsory from each question. Attempt any* ***one*** *part from (b) & (c).*

1. (a) Define Piecewise continuous trial function for Rayleigh-Ritz method. 2

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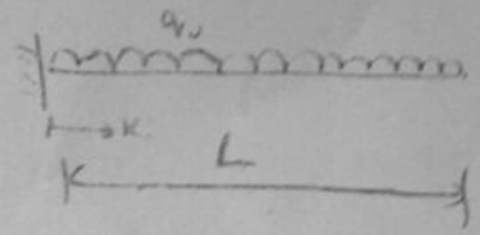
(b) Find the trial solution for a cantilever beam under uniform distributed load q_0 as shown.



Fig

The of Governing differential equation is given by

$$EI\frac{d^4v}{dx^4} - q_0 = 0$$

and the boundry conditions are given by

$$v(0) = 0;\ \frac{dv}{dx}(0) = 0,\ \frac{d^2v}{dx^2}(L) = 0,$$

$$\frac{d^3v}{dx^3}(L) = 0.$$ 14

(c) Obtain Rayleigh-Ritz solution for a simply supported beam under UDL q_0 as shown

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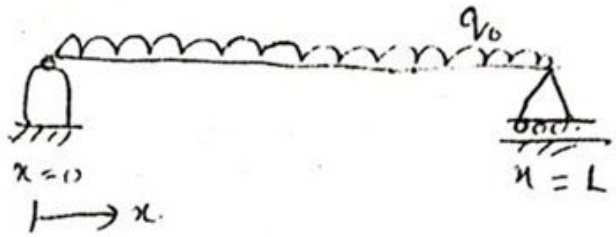

$$EI\frac{d^2v}{dx^2} - q_0 = 0$$

and Boundry conditions are

$$v(0) = 0;\ v(L) = 0,\ \frac{d^2v}{dx^2}(0) = 0,$$

$$\frac{d^2v}{dx^2}(L) = 0.$$

2. (a) Define shape function. 2

(b) Derive the shape function of a two noded linear bar element. 14

(c) Derive the shape function of a quadratic bar element of length and having three nodes. 14

3. (a) Define Beam and Frame element. 2

(b) Derive the stiffness for beam element. 14

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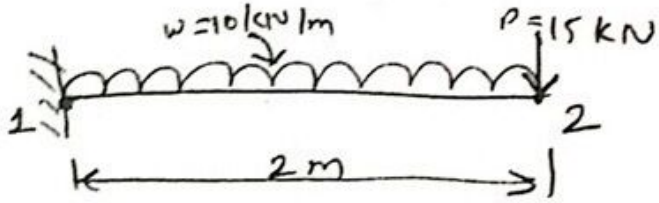
(c) Find the transverse displacement and rotation displacement at point 2 of a cantilever beam of 2 m span loaded with UDL and point load as shown. 14



Take $E = 210$ GPa, $I = 4 \times 10^{-4}$ m^4,

$W = 10$ kN/m, $P = 15$ kN.

4. (a) Write shape functions for simple three-node triangular element. http://www.csvtuonline.com 2

(b) Determine the shape function of a four noded rectangular element. 14

(c) Explain in brief Gauss Quadrature technique? Evaluate

$$I = \int_{-1}^{1} \left(2 + x + x^2\right) dx.$$ 7+7

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5. (a) Write the eqation for plain stress condition in matrix form. 2

(b) Explain finite element analysis for plane strain. Derive relationship between stress-strain equation for plane stress condition. 7+7

(c) Derive Strain Displacement matrix for 2 D element. 14

(2D)

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